LOGICSHIELD

AI-POWERED DEBATE SIMULATION & COMMUNICATION RISK FORECASTING

Adversarial Argument Testing, Fallacy Mining & Structured NLP Risk Assessment

TECHNICAL BRIEF
Version 2.0 (Stable)
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LogicShield is an advanced AI-powered Communication Intelligence Platform. It bridges adversarial debate training with structured Natural Language Processing (NLP) risk forecasting. Designed to strengthen argument construction, detect logical fallacies, and evaluate potential reputational risk, the platform provides real-time coaching as users compose their texts. LogicShield evaluates coherence, sentiment balance, and language sensitivity, giving communication professionals, students, and executives a sandbox to stress-test their ideas before they perform, pitch, or publish.

SYSTEM CORE ARCHITECTURE

LogicShield is engineered with a modern, decouple-oriented architecture that optimizes processing latency for real-time analysis:

- **Next.js 14 Frontend:** Built with React 18, TypeScript, and Tailwind CSS. The Client UI features debounced asynchronous inputs (500ms trigger) to dispatch content drafts to the API.
- **FastAPI (Python) Backend:** An asynchronous ASGI high-performance framework. It exposes REST endpoints, orchestrates ORM operations, manages user sessions, and houses the NLP model service pipelines.
- **SQLAlchemy 2.0 ORM:** Connects to a PostgreSQL database for enterprise production environments and SQLite for local development structures.
- **JWT-Based Auth:** Secure JSON Web Token authentication with python-jose, handling profile management and learning telemetry persistence.

ADVERSARIAL DEBATE SIMULATOR

The platform provides an active, turn-based dialogue simulator where users debate an AI opponent to test their argumentative stability:

- **Configurable Personas:** Opponents are set with distinct reasoning types: *Logical* (evidentiary support), *Aggressive* (direct refutation), *Skeptical* (hyper-critical inquiry), and *Devil's Advocate* (alternative perspective framing).
- **Contextual Counter-Arguments:** Powered by Meta Llama 3.2 1B (meta-llama/Llama-3.2-1B-Instruct) via Hugging Face Inference endpoints or local Ollama servers. The LLM processes the history, persona profiles, and user stances to generate highly relevant rebuttals.
- **Voice-Activated Mode:** Supports hands-free debates using the native Web Speech API for speech-to-text input, and Speak-TTS for synthesized voice output.

NLP & ML ANALYSIS PIPELINE

The analysis pipeline compiles data from multiple specialized transformer models to output real-time scoring metrics:

Service Domain	Underlying NLP Model	Target Metrics & Role
Fallacy Classify	BART-large-MNLI	Zero-shot multi-label classification across 9 types.
Coherence Scoring	all-MiniLM-L6-v2	Sentence-BERT semantic embedding cosine similarity.
Sentiment Analysis	DistilBERT-SST-2	Polarity and emotional intensity estimation.
Toxicity Risk	toxic-comment-model	Estimates inflammatory and offensive phrasing score.
Hate Speech Scan	roberta-hate-speech	Detects identity-sensitive and discriminatory claims.

LOGICAL FALLACY CLASSIFIER

The platform automatically flags errors in reasoning as users type, supporting 9 common patterns:

- **Supported Fallacies:** *Ad Hominem* (attacks source), *Straw Man* (misrepresentation), *False Dilemma* (binary limitation), *Slippery Slope* (unsubstantiated progression), *Appeal to Authority* (unqualified sourcing), *Bandwagon* (popularity appeal), *Circular Reasoning* (premise is conclusion), and *Red Herring* (diversion).
- **Detection Mechanics:** Utilizes multi-label zero-shot classification to map text vectors against dynamic hypothesis templates. Includes a rule-based fallback keyword scanner to handle offline configurations.

REPUTATION RISK EVALUATION

A B2B-oriented diagnostic engine that scores drafts to forecast public backlash and corporate communication risks:

- **Aggregate Risk Score:** Evaluated across five weighted categories: Identity-Sensitive language (30%), Inflammatory phrasing (25%), Absolutist vocabulary (20%), Moral polarity (15%), and Defensive posture (10%).
- **Risk Categorization:** Scores mapped into *Low* (<0.2), *Medium* (0.2-0.5), *High* (0.5-0.75), and *Critical* (>0.75) backlash tiers with inline highlighting and automatic rewrite options.

INTERACTIVE RHETORIC ACADEMY

The platform features a slide-based course player containing 6 fully structured courses comprising 14 interactive lessons, designed to train users in formal argumentation theory:

- **Animated Widgets:** Lessons integrate custom interactive widgets: *Syllogism Builders* (deductive chains), *Fallacy Spotters* (dialogue analysis), *Causal Confounders* (correlation vs. causation models), and *Reframe Balance Scales* (objection resolution).
- **Exam & PDF Certificates:** Lessons conclude with concept quizzes. Achieving an 80% passing grade on the comprehensive course exams triggers the automated creation and rendering of PDF completion certificates.

PERSONALIZED LEARNING PATH

A dynamic learning engine that analyzes individual skill levels and structures adaptive reviews:

- **Onboarding Diagnostics:** A 5-question logic assessment that classifies user proficiency (Beginner, Intermediate, Advanced) and identifies specific cognitive weaknesses.
- **Spaced Repetition System (SRS):** Automatically schedules interactive flashcards focusing on identified weak categories to enhance retention.
- **Reactive Path Gate:** The Course Academy is locked until onboarding completion. The system issues custom window events (learning-progress-updated) to trigger immediate, non-refresh UI state updates.

SYSTEM DEPLOYMENT CONFIGURATIONS

The application is optimized for dual deployment scenarios depending on compute availability and hardware configurations:

- **Demo Mode:** Enabled via env configurations (DEMO_MODE=true). Restructures endpoints to utilize regex templates and local keyword dictionaries. Bypasses model downloads (~4GB VRAM/RAM) to function immediately out-of-the-box.
- **Full ML Pipeline:** Runs PyTorch-based transformers locally on GPU/CPU for offline operation, or interfaces with cloud inference servers.

RELATIONAL DATABASE ARCHITECTURE

The SQLAlchemy ORM maps relational database tables. Core tables enforce standard integrity constraints and link elements across sessions:

- **users:** Account records with unique index fields on emails and usernames. Encrypts passwords via Bcrypt and maintains activation flags.
- **debate_sessions:** Stance configuration mappings (support, oppose, neutral), topics, and selected AI personas.
- **arguments:** Historically captures the debate transcript with session mappings and sender identifiers.
- **analysis_results:** Analysis outcomes linked 1:1 with arguments. Maps JSON vectors of fallacy categories, numeric strength outputs, and string risk levels.
- **user_analytics:** Aggregate stats record per user: tracks total sessions, rolling average strength, and computed weak fallacies.

COMPETITIVE DOCKING MATRIX

A feature evaluation of LogicShield against leading debate, argument mapping, and communication risk management platforms:

Core Feature	LogicShield	Kialo	Debatewise	Talkwalker
Real-Time Suggestions	**Yes (Live)**	No (Post-submit)	No	No
Fallacy Classify	**Yes (ML)**	No	No	No
Rhetoric Academy	**Yes (Interactive)**	No	No	No
Reputation Risk	**Yes (ML)**	No	No	Yes (Social)

PLATFORM METRICS & STATUS

- **Development Phase:** Phase 2 Active. Course Academy, Speech Mode, and Onboarding Path are fully completed.
- **Prerequisites:** Modern Web Browser. Backend supports Python 3.10+ running FastAPI on standard hardware (supports optional GPU acceleration).
- **Repository Specs:** Includes modular directories for web/ (Next.js) and backend/ (FastAPI) ensuring streamlined microservice deployments.